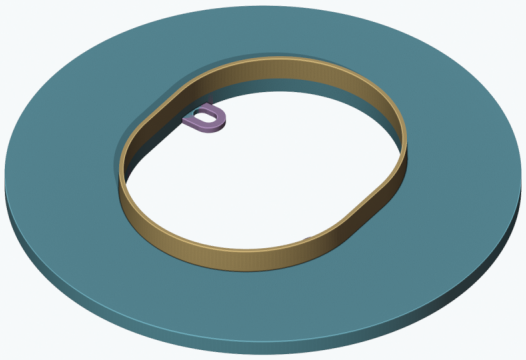
Print this one

For the friend operating the printer | One part, one material

Open PRINT_ONE_RF03_D12.3mf. If unsupported, use the matching STL in millimeters. Both are the same part; print one copy. The 3MF carries geometry only - keep your own printer profile.

PRINT ONE: RF03 / 12 mm continuous collar



240 mm across | 8 mm face | 2 mm collar wall | 12 x 8 mm slots

Fit on the bed before starting

240 x 240 x 20 mm model. Allow **250 x 250 mm usable area** with a 5 mm brim; check clips, purge lines and exclusion zones. A 220 mm bed cannot fit it. Keep **100% scale**; do not auto-shrink. If it does not fit, request a split design.

Use these starting settings

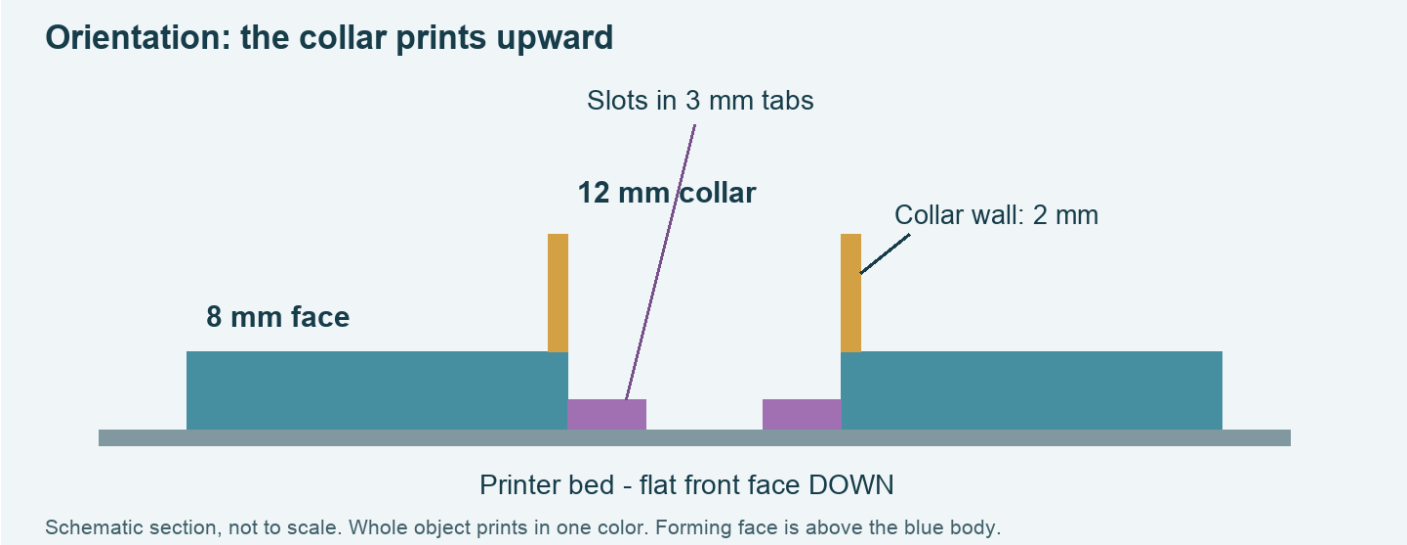
Unfilled PETG (one color), 0.4 mm nozzle, 0.20 mm layers, **5 walls**, **1.2 mm top and bottom skins**, **25% gyroid/cubic infill**, 5 mm brim. Supports OFF; raft OFF. Use the printer/filament maker's established PETG profile and bed-release advice.

Files already sit flat, collar upward. Preview the layers: solid skins, intact 12 x 8 mm slots, collar made from continuous walls. No G-code is supplied because machine settings are unknown.

After printing: remove brim and strings, clear only first-layer squeeze at slots, check the face against a flat surface and hand the whole print back to the owner. Avoid bending the tabs.

Use & orientation

For the owner / installer | A removable tool, not the final escutcheon



Keep the simple collar as the first choice

D12: 12 mm continuous collar, 20 mm overall including the 8 mm face.
Optional D24_RELIEVED: 24 mm maximum depth, 32 mm overall. Two rear cutaways clear captured obstacles. Only the first 12 mm remains a continuous molding wall. Choose it instead of D12 only if deeper guidance is useful; rehearse clearance and removal.

Before any fill

1. Remove the rod-end nuts and trim as required. Protect the valve and stone. Locate loosely using the two smaller slots; never force the form over the nuts.
2. Rehearse full-face seating and straight withdrawal toward the shower. The collar checks do not prove the broad face will sit flat on real stone.
3. Use broad, padded independent bracing on the front annulus. Rods are alignment aids only. Do not tighten their nuts to carry repair load or pull a warped print flat.

Repair and removal

Use only with sound permanent backing and the installer-selected compatible repair system. Line with removable approximately 0.05 mm PE release film. Hand-place only the needed supported perimeter zones within the front 12 mm depth, outside the collar. Keep the central service space, rods and tabs free of fill. No pressure casting or hard tamping.

After the repair reaches suitable stripping strength, remove the brace and withdraw the tool. Remove every plastic piece, film and pad. Inspect the repair, then dry-fit the actual trim/plates and resolve the separate surface sealing and curing detail. Nothing here is approved as a permanent waterproof liner.

The thinner face relies on the same independent support as RF02. Print durability, actual seating and release have not been physically tested. Stop if it rocks, bows or catches; do not alter the travertine to make the tool fit.

Budget & file choices

For the owner | Estimate in US dollars, not a vendor or labor quote

A generous payment: $75 for one selected form

This budgets a whole spool, ordinary setup/cleanup, power, wear and one failed attempt. It is intentionally more than the consumed-plastic cost. It assumes a hobby favor, not commercial shop labor or a new spool dryer/printer purchase.

Whole 1 kg PETG spool, delivery/tax allowance	**$45**
Electricity, drying and one retry allowance	**$5**
Wear and consumables allowance	**$10**
Setup / cleanup thank-you allowance	**$15**

Material calculation

D12 planning estimate: **160-240 g including routine brim/purge** with the stated settings. At the conservative $35/kg budget that is $5.60-$8.40 in plastic. Even a completely solid CAD body is about **346 g** before brim/purge. The optional deep collar adds only about 11 g if printed solid.

These are geometry-based estimates, not an actual slicer report. Budget roughly 6-16 print hours per attempt; printer speed can move this substantially. Power assumption: 0.20 kW average x 16 h x $0.50/kWh = $1.60, plus drying/heat-up. A $5 allowance covers two such prints and about 1.5 kWh extra drying/heat-up. These are conservative budget inputs, not measured electricity use or a local utility tariff.

Which files to send

Send the entire RF03_FRIEND_PRINT_PACKAGE.zip. Root 3MF/STL = default D12. OPTIONAL_DEEPER = print-instead alternative. CAD_REFERENCE = editable STEP equivalents, each one tool solid, in the same millimeter print pose. PREVIEW and ORIENTATION identify the shape. No scans, home photos, wall assembly, factory parts or printer commands are included.

Sources checked 17 October 2026

Prusa standard PETG category displayed $25.49/kg sale listings; the budget uses $35/kg, not that sale or a checkout quote. Polymaker's PETG technical sheet lists 1.30 g/cm3. Prusa's PETG and infill guides support using suitable bed preparation and walls/infill rather than blindly making a solid print. Exact URLs and assumptions are in COST_ESTIMATE.json.

Prusa PETG catalog

Polymaker PETG technical sheet

Prusa PETG guide

Prusa infill guide